

SDTH 3 Month

Propose your venture by June 13

Our guidelines

1 First Impressions

Team name, logo, and a two-liner on the problem and core ambition.

2 Team

Names, faces, strengths, beliefs, and why this problem matters to you.

3 Scenario

The operating context: where the problem appears, who faces it, and why it matters here.

4 Shortfalls of the Status Quo

The constraints that make current approaches too slow, brittle, costly, or hard to deploy.

5 Proposed Solution

One clear statement on your product and the gap it closes.

6 Core Capabilities

The 3–4 things the system must do well to be useful.

7 Why this differs

The design choices that make your approach meaningfully different from existing ones.

8 3 Month Plan

The phases, deliverables, and proof points you aim to reach by the final demo.

9 Risks and Mitigations

The main technical, operational, or integration risks, and how you intend to reduce them.

10 Support and Integration

The mentors, data, hardware, test access, or partner teams that would accelerate the work.

11 Closing Argument

What becomes possible if the project succeeds beyond the hackathon.

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Frame the project

1 First Impressions

Team name, logo, project title, and a two-liner on the problem and core ambition. The reader should understand the project before turning the page.

Example: *Akro - Agentic C-UAS Command and Control - Software is the fastest on the battlefield. Let it be ours.*

2 Team

Names, faces, relevant background, and likely roles. Show why this team has the strengths, beliefs, or lived context to take on the problem.

Example: *A short line per person is enough: role, relevant experience, and what they own in the build.*

3 Scenario

The operating context: where the problem appears, who faces it, why it matters here. Use concrete Singapore-specific constraints where possible.

Example: *Group 1-3 UAS launched cheaply, in numbers, against a dense island airspace already carrying heavy commercial traffic.*

Useful details	Examples
Geography / density	Island scale, urban canyons, port approaches, crowded airspace.
System constraints	Latency, power, weight, range, cost, comms loss, sensor limits.
Users / platforms	Soldiers, operators, drones, robots, sensors, command layers.

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Define the build

4 Shortfalls of the Status Quo

Name the constraints that make current approaches too slow, brittle, costly, or hard to deploy. Explain why the obvious solution is not enough.

Example failure categories

Sensor heterogeneity

EO, IR, RF and acoustic sensors speak different protocols. No common fusion layer exists in fielded systems.

Decision time

Human-in-the-loop approval cycles can run in minutes. Threat timelines may be far shorter.

Vendor lock-in

Proprietary stacks prevent cross-vendor interoperability. Updates require vendor access, not operator action.

Doctrine mismatch

Commercial autonomy stacks are not built around ROE. Fielding them requires re-engineering, not simple integration.

5 Proposed Solution

One clean statement on your product and the gap it closes. Include who it helps, what it does, and the constraint it is designed for.

Example: *Red Dome closes the sensor-shooter loop from minutes to seconds.*

6 Core Capabilities

The 3-4 things the system must do well to be useful. For each: what it takes in, what it outputs, and why it matters operationally.

Example capability blocks

Sensor fusion

Ingests EO/IR, RF, or acoustic streams via open adapters. Normalises inputs to a common track format.

Threat classification

Classifies threats by behaviour and signature. Outputs confidence with an evidence chain.

COA recommendation

Ranks options by ROE compliance, collateral risk, and engagement geometry.

Effector tasking

Sends commands to kinetic or non-kinetic effectors without proprietary protocol dependency.

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Make the case

7 Why this differs

The design choices that make your approach meaningfully different from existing ones. Show why a startup-speed team should be trying this.

Example design choices

Open protocols

Interfaces are open or documented.
Operator-owned systems can connect without vendor involvement.

Doctrine-aware

ROE constraints are first-class inputs, not post-hoc filters.

Edge-first

Runs on COTS edge compute with degraded comms. No cloud dependency or SaaS call-home.

Built for Singapore

Designed around Singapore geography, density, threat profiles, and operating constraints.

8 3 Month Plan

The phases, deliverables, and proof points you aim to reach by the final demo. Where possible, define one measurable outcome that shows meaningful progress.

Example phased plan

	Activities	Deliverables	Success criteria
Build Weeks 1-4	Core adapter, baseline model, first data ingestion.	Working module or prototype with documented build.	Threshold met on replay or bench test. Pipeline ready for live input.
Integrate Weeks 5-8	Fusion, classification, interface work, early user feedback.	Integrated module with confidence scores, logs, or test harness.	Measured latency / accuracy / power on agreed test set.
Demo Weeks 9-12	Scenario rehearsal, partner integration, final evidence package.	Live or recorded end-to-end demo with documentation.	Target reproduced in front of mentors / sponsors, with logs.

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Ask and close

9 Risks and Mitigations

Name the main technical, operational, resource, or integration risks. For each, state how you intend to reduce, bypass, or test the risk early.

Example risk table

Risk	Category	Mitigation
Partner interface spec slips past Week 6.	Integration	Begin with a mocked interface; switch to live integration after spec freeze.
Drone test slot constrained by airspace or range access.	Operational	Pre-book early. Design replay or simulation fallback that still tests the success criterion.
Classifier degrades on novel signatures.	Technical	Reserve time for adversarial testing. Define acceptable degradation floor in advance.
Key member unavailable during integration week.	Resource	Document hand-offs after each phase so no one person becomes a single point of failure.

10 Support and Integration

What would accelerate the work: mentors, data, hardware, test access, operator feedback, technical partners, or other SDTH teams. Explicit asks make sponsors and mentors useful.

Example asks

From organisers	From partner teams
Doctrine or ROE references, datasets, simulation environments, sensor logs, hardware access, test sites, operator feedback.	Interface specifications, data schemas, model outputs, integration windows, joint demo plan.

11 Closing Argument

What becomes possible if the project succeeds beyond the hackathon. What is true after three months that is not true today?

Example: End by making clear why this work deserves follow-on support.